

10 MICRO SERVICES PROJECT

( GitHub - usubbu/microservices-project)

STEP-1: LAUNCH T2.LARGE INSTANCE WITH ADMIN PERMISSIONS

STEP-2: Install AWS CLI, kubectl, and eksctl

Install AWS CLI

```
curl "https://awscli.amazonaws.com/awscli-exe-linux-x86_64.zip" -o "awscliv2.zip"
```

```
unzip awscliv2.zip
```

```
sudo ./aws/install
```

TO SET PATH: vim .bashrc

```
export PATH=$PATH:/usr/local/bin/
```

```
source .bashrc
```

Install KUBECTL:

```
curl -o kubectl https://amazon-eks.s3.us-west-2.amazonaws.com/1.19.6/2021-01-05/bin/linux/amd64/kubectl
```

```
chmod +x ./kubectl
```

```
sudo mv ./kubectl /usr/local/bin
```

```
kubectl version --short --client
```

Install EKCTL:

```
curl --silent --location "https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_$(uname -s)_amd64.tar.gz" | tar xz -C /tmp
```

```
sudo mv /tmp/eksctl /usr/local/bin
```

```
eksctl version
```

STEP-3: Create EKS Cluster

create cluster:

```
eksctl create cluster --name=EKS-1 --region=ap-south-1 --zones=ap-south-1a,ap-south-1b --without-nodegroup
```

Attach IAM Role:

```
eksctl utils associate-iam-oidc-provider --region ap-south-1 --cluster EKS-1 --approve
```

create NodeGroup:

```
eksctl create nodegroup --cluster=EKS-1 --region=ap-south-1 --name=node2 --node-type=t3.medium --nodes=3 --nodes-min=2 --nodes-max=4 --node-volume-size=20 --ssh-access --ssh-public-key=mustafakey-pair --managed --asg-access --external-dns-access --full-ecr-access --appmesh-access --alb-ingress-access
```

UPDATE CLUSTER:

```
aws eks update-kubeconfig --name EKS-1 --region ap-south-1
```

STEP-4: Install Jenkins & Docker

- `sudo wget -O /etc/yum.repos.d/jenkins.repo https://pkg.jenkins.io/redhat-stable/jenkins.repo`
- `sudo rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key`
- `yum install java-17-amazon-corretto -y`
- `yum install jenkins -y`
- `systemctl start jenkins.service`
- `systemctl status jenkins.service`
- `yum install docker -y`
- `systemctl start docker`
- `chmod 777 ///var/run/docker.sock`

STEP-5: Install Plugins: Install the following Jenkins plugins:

- Docker Pipeline
- Kubernetes
- Kubernetes CLI

STEP-6: Now add the dockerhub Credentials

STEP-7 Create name space & Service Account

Namespace: `kubectl create ns webapps`

ServiceAccount:

apiVersion: v1

kind: ServiceAccount

metadata:

name: jenkins

namespace: webapps

Create Role:

apiVersion: rbac.authorization.k8s.io/v1

kind: Role

metadata:

name: app-role

namespace: webapps

rules:

- apiGroups:

- ""

- apps

- autoscaling

- batch

- extensions

- policy

- rbac.authorization.k8s.io

resources:

- pods

- componentstatuses

- configmaps

- daemonsets

- deployments

- events

- endpoints

- horizontalpodautoscalers

- ingress

- jobs

- limitranges

- namespaces

- nodes

- pods

- persistentvolumes

- persistentvolumeclaims

- resourcequotas

- replicaset

- replicationcontrollers

- serviceaccounts

- services

verbs: ["get", "list", "watch", "create", "update", "patch", "delete"]

Bind the role to service account:

apiVersion: rbac.authorization.k8s.io/v1

kind: RoleBinding

metadata:

name: app-rolebinding

namespace: webapps

roleRef:

apiGroup: rbac.authorization.k8s.io

kind: Role

name: app-role

subjects:

- namespace: webapps

kind: ServiceAccount

name: jenkins

Generate token using service account in the namespace:

apiVersion: v1

kind: Secret

type: kubernetes.io/service-account-token

metadata:

name: mysecretname

annotations:

kubernetes.io/service-account.name: jenkins

Now it will generate a token, Copy this token and create the credential in jenkins named k8-token.

Go to credentials >> select secret text >> copy paste it and id as **k8s-token**

STEP-7: Set Up Multibranch Pipeline

Add this Jenkins file on your github repo

```
pipeline {
```

```
    agent any
```

```
    stages {
```

```

stage('Deploy To Kubernetes') {

    steps {

        withKubeCredentials(kubect credentials: [[caCertificate: '', clusterName: 'EKS-1',
contextName: '', credentialsId: 'k8-token', namespace: 'webapps', serverUrl: 'add-your-eks-cluster-url']]) {

            sh "kubectl apply -f deployment-service.yml"

        }

    }

}

stage('verify Deployment') {

    steps {

        withKubeCredentials(kubect credentials: [[caCertificate: '', clusterName: 'EKS-1',
contextName: '', credentialsId: 'k8-token', namespace: 'add-your-eks-cluster-url']]) {

            sh "kubectl get svc -n webapps"

        }

    }

}

}

```

command to delete cluster : eksctl delete cluster --name EKS-1 --region ap-south-1